

Summary of Remedial Options: North and South Waste Rock Piles

Former Sunro Mine

October 22, 2020



Teck

Presentation Outline

Introductions

Summary of Recent Work

Remedial Options Analysis

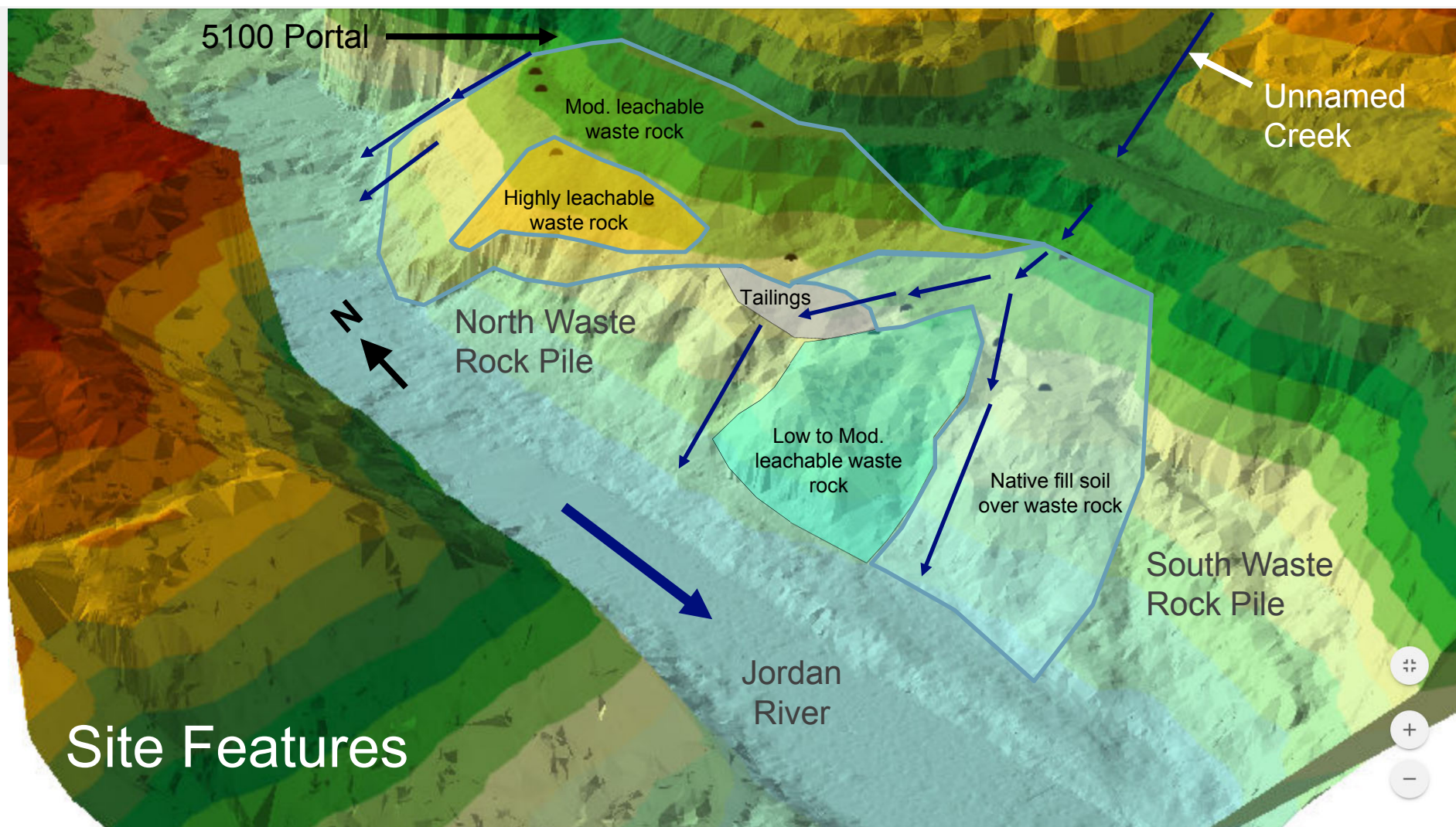
Next Steps and Schedule

Open Discussion



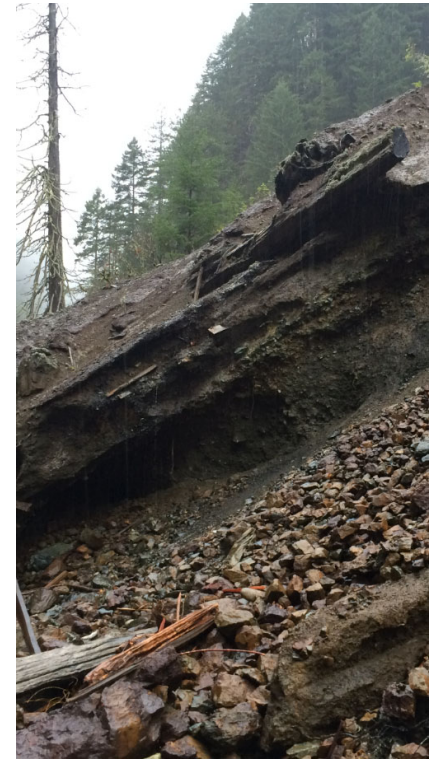
Recap of Project Objectives

- Assess potential contamination in water and season effects.
- Determine contaminant loading to the Jordan River.
- Evaluate the potential aquatic risk.
- Assess options to address areas where risks are predicted and/or adverse conditions are observed.
- Comply with the ENV order to address Protocol 12 “high- risk” conditions.
- In 2019, decided to proceed with Remediation of Options analysis of Waste Rock while gathering more information on portal discharge.



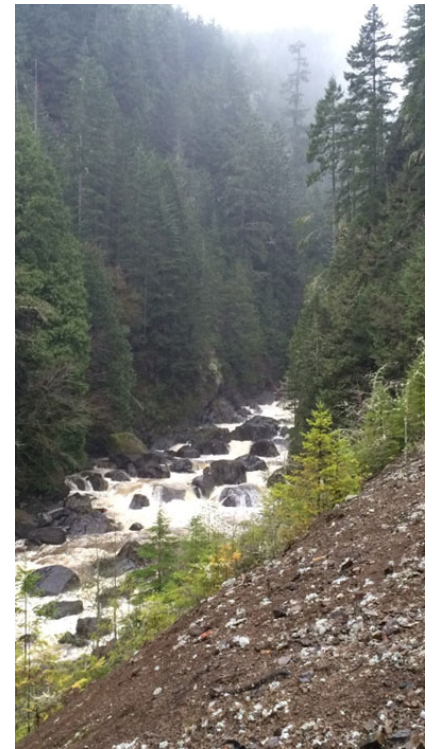
Summary of Key Considerations: Soil

- High sulphide, highly leachable and acid generating waste rock and tailings ($\sim 30,000 \text{ m}^3$).
- Low sulphide, moderate to low leachable and not acid generating waste rock mixed with native soil ($\sim 60,000 \text{ m}^3$).
- Low metal concentrations in soil from a Human Health perspective.
- Eroding toe and unstable slopes of the North Waste Rock Pile.



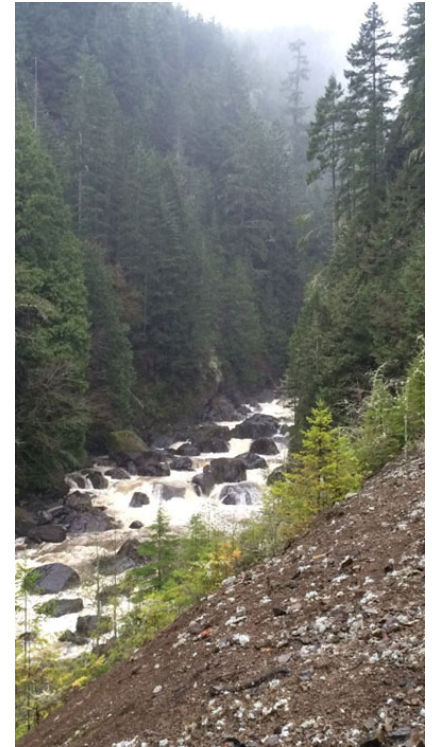
Summary of Key Considerations: Water

- 5100 Portal discharge is contaminated with copper and other metals; studies are ongoing.
- Unnamed Creek #1 and #2 contribute very little copper loading via stream discharge. Creek may infiltrate and contribute to groundwater contaminate loading to the Jordan River.
- The Site contributed to copper concentrations in the Jordan River during all sampling events.

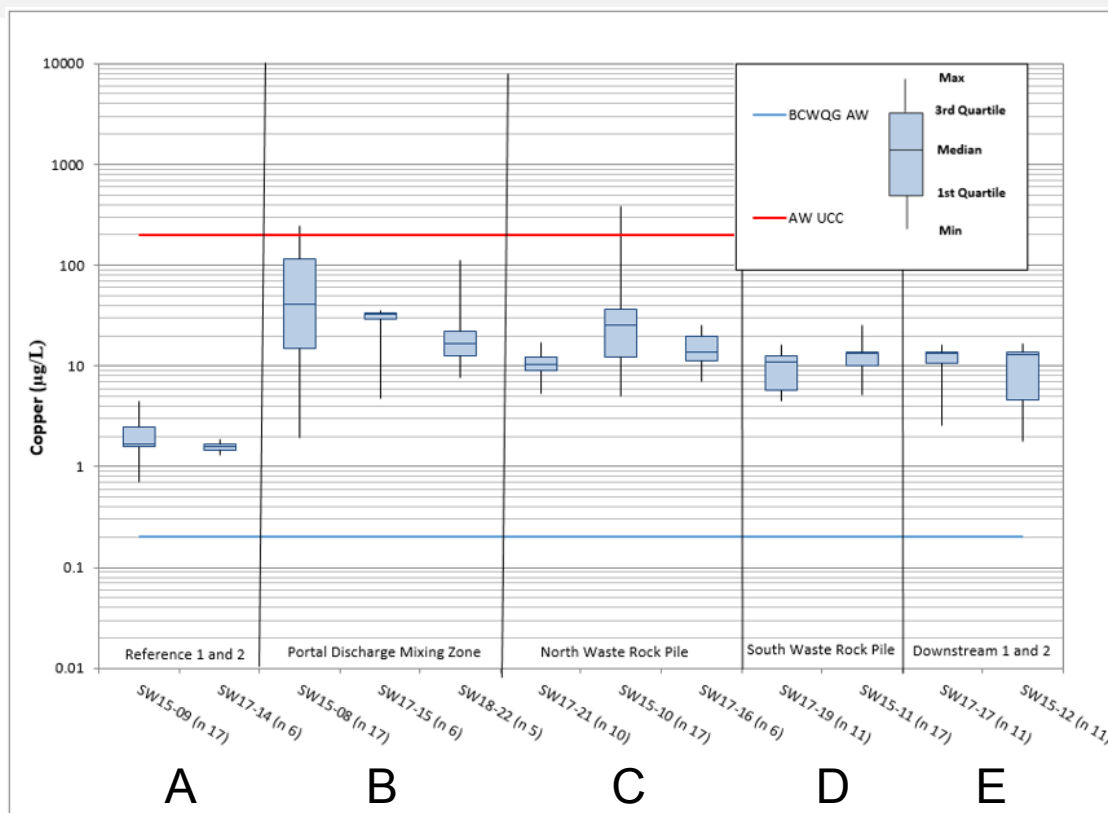
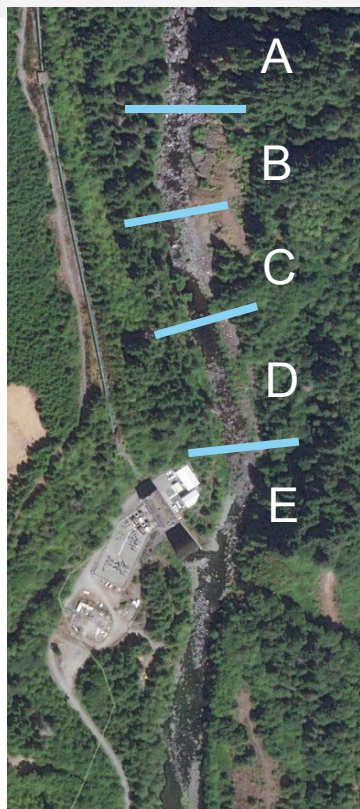


Summary of Key Considerations: Water

- The Portal is a major source of copper loading to the river during dry periods (~ 90% of total load).
- The North Waste Rock Pile flush metals into the river during heavy rain events (~ 60% of total load).
- The South Waste Rock Pile does not appear to be major source of metal loading to the Jordan River.

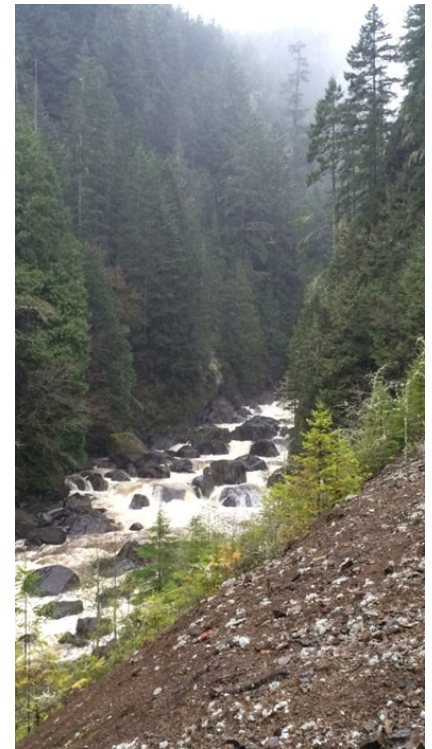


Statistical Summary of Dissolved Copper in the Jordan River: Surface Water



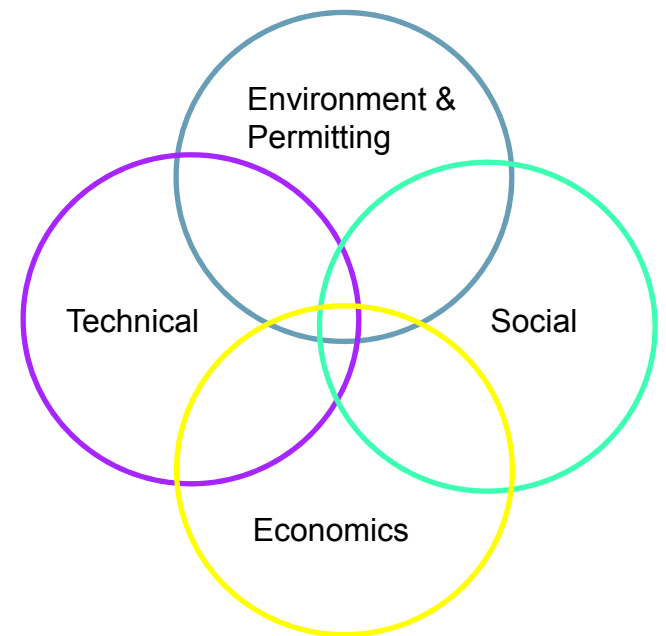
Summary of Key Considerations: Water

- Field study of benthic invertebrates did not identify copper-related effects to communities with some uncertainty to sensitive EPT species.
- Risks to fish in the Jordan River were considered low to moderate with some uncertainty.
- The copper exposure gradient is confounded by differences in habitat and other factors, making it hard to determine site discharges affecting the fish community in the lower Jordan River.
- Physical remediation of soil and water is required to address Protocol 12 “high-risk” conditions.



Multiple Accounts Analysis: What is it?

- A tool to assist in a decision-making process.
- Uses agreed upon accounts, remedial objectives, sub-objectives and measurement criteria.
- Transparent value-based decision process with scoring and weighting in a workshop setting.
- Note: that the highest ranking score does not necessarily mean that option should be chosen as the preferred remedial option.



Multiple Accounts Analysis – Methodology

Steps	Name	Purpose
1	Identify Remedial Alternatives	Develop a list of all possible candidate remedial alternatives.
2	Pre-screening Assessment	Fatal flaw analysis to drop alternatives that do not meet the minimum requirements for remediation.
3	Alternative Characterization	Development of Site specific evaluation of value groupings (“Accounts”): technical; economics; environmental and permitting; technical; and, social.
4	Multiple Accounts Ledger	Development of remedial objectives, sub-objectives and measurement criteria used in Step 5 and 6.
5	Value-Based Decision Process	Assign scoring and weightings in the Tool for the value groupings, remedial objectives and measurement criteria.
6	Sensitivity Analysis	Vary weightings to assess bias and subjectivity in the evaluation team.

Multiple Accounts Analysis – Scoring and Weighting

- The degree to which a particular remedial option meets an objective as determined by the sub-objective is given a qualitative score between 1 (worst) and 9 (best).
- Scores of 1 and 9 were always applied to each sub-objective and the other options were scaled in between the best and worst options.

Initial Score	Scoring for Sub-Objectives
1	Worst Remedial Option (e.g., Does not Meet Objectives / Very High Costs)
2 - 8	Remaining option scaled between worst and best options and scored based on perceived ability to meet objectives
9	Best Remedial Option (e.g., Fully or Partially Meets Objectives / Low Costs)

Multiple Accounts Analysis: Three Workshops

Opportunity Framing Workshop (OFW) on April 24-25, 2017 in Kimberley, BC .

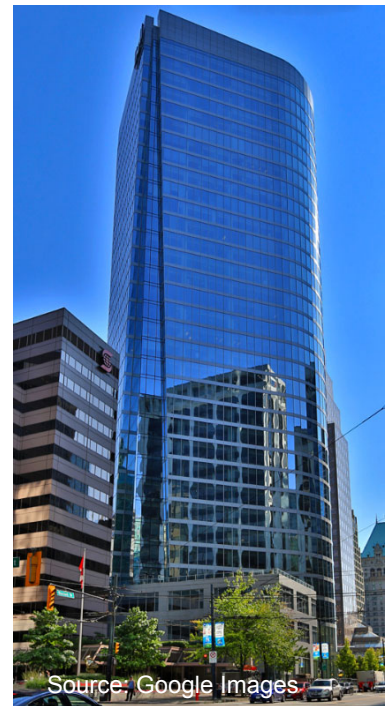
At the OFW the participants identified candidate alternatives (Step 1) and completed the pre-screening assessment (Step 2).

Initial Workshop on May 8, 2019 in Vancouver, BC.

At the Initial Workshop the participants completed the alternative characterization (Step 3) and populated the multiple accounts ledger (Step 4).

Final Workshop on December 10, 2019 in Kimberley, BC.

During the workshop the attendees completed the value-based decision process (Step 5) and completed a first draft of the sensitivity analysis (Step 6).



Accounts, Objectives and Measures

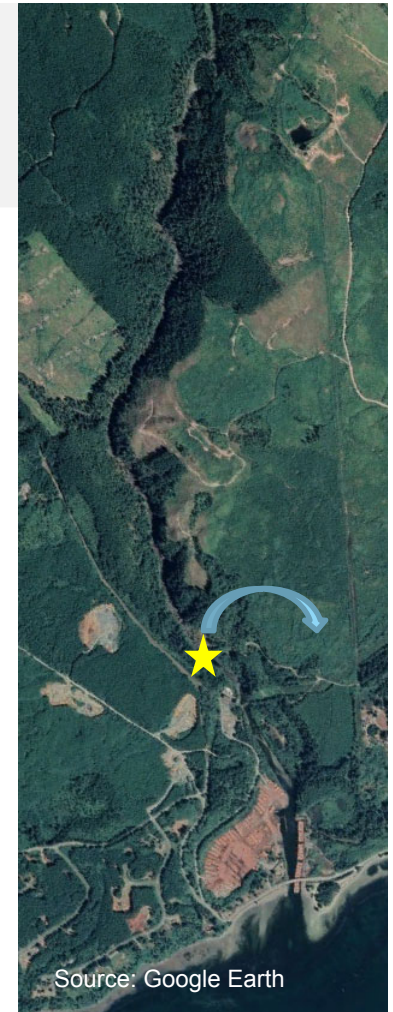
Accounts	Objectives	Sub-Objectives	Measures (Considerations)
Technical	Maximize physical safety during construction	Worker safety	Incident frequency and severity
		Public safety	No public incidents
	Minimize ecological impacts from mine waste to level acceptable under CSR	Aquatic life (fish, invert, etc.)	Likelihood of residual (post-remediation) risk being acceptable
		Terrestrial	Likelihood of residual (post-remediation) risk being acceptable
	Minimize environmental impacts arising during remediation activities	Adverse carbon footprint impacts minimized	Quantity of CO2 emissions
		Risk to water quality	Likelihood of shock loads to Jordan River
		Risk to air quality/noise due to fugitive dust from trucks	Reduced noise and dusting
	Minimize engineering risk	Ease of constructability	Less engineering safety measures needed
		Ability to handle climatic variability	Design can accommodate variability without additional infrastructure or active oversight
	Minimize routine post-remediation monitoring and maintenance requirements	Level of monitoring and maintenance effort	Person-days/year of effort (including on-site and office reporting), level of technical skills required for on-site activities
	Maximize resilience of physical remediated site	Seasonal high flows/flooding	Water flows controlled, and Soil and water chemistry remains below guidelines, no geotechnical hazard
		Climate change	No additional infrastructure needed
		Seismic events	Earthquakes/tsunamis
		Human-caused effects	Water flows remain controlled

Accounts, Objectives and Measures

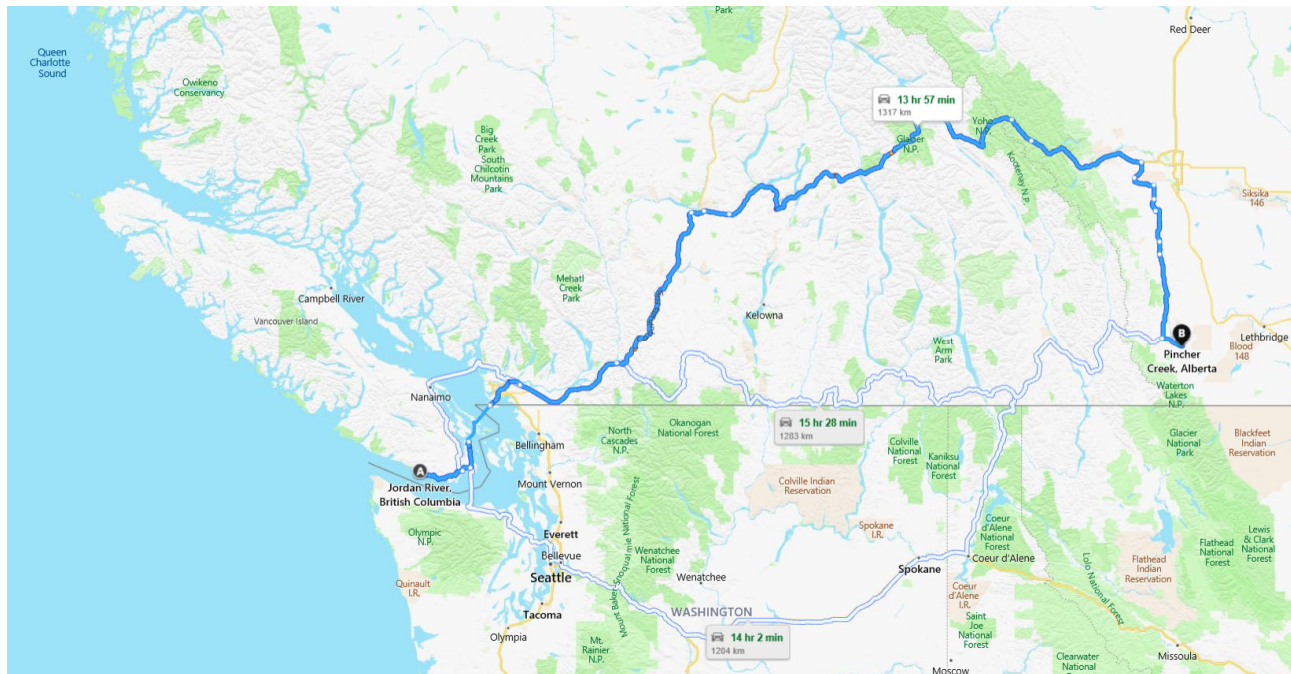
Accounts	Objectives	Sub-Objectives	Measures (Considerations)
Economics	Minimize total cost	Initial CapEx	\$ million
		Budget Risk	\$ million
		OpEx	\$/yr
Enviro & permitting	Maximize incremental environmental benefits from remediation	Aquatic habitat value improves	Additional habitat features constructed
		Terrestrial habitat value improves	Additional habitat features constructed
	Minimize future environmental liability	Minimize likelihood of increased scope of responsibilities	Liability to Teck
		Minimize likelihood of future remediation requirements	Liability to Teck due to potential future changing standards
	Obtain permits to conduct remediation	Minimize permitting complexity & effort	Amount and duration of studies req'd to support permit appl
		Maximize permitting certainty	Likelihood of obtaining approvals
	Regulatory requirements are met	Maximize regulatory (MOE) support	High risk conditions are addressed to MOE satisfaction
	Maximize acceptability to other land/tenure holders	Western Forest Products Landowner	Third party acceptance
		New Sunro underground mine owner	Third party acceptance
Social	Maximize social acceptance	Salmon Restoration Society	Agreement on preferred remedial plan
		Jordan River community & Immediate neighbours	
		T'sou-ke	
		Pacheedaht First Nation	

Candidate Options: Waste Rock and Tailings

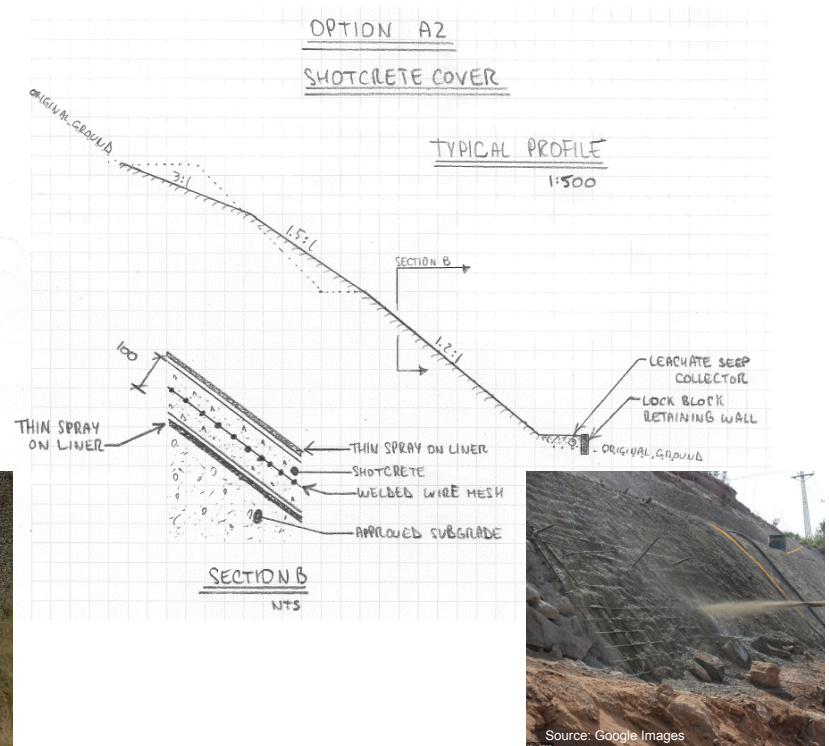
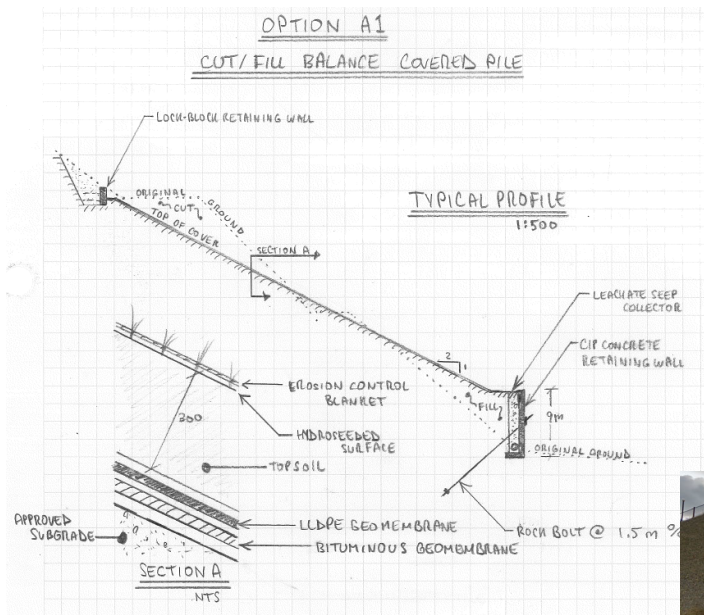
- Physically stabilize all waste rock in-situ and control seepage by capping.
- Move all waste rock to an on-site or off-site landfill. Onsite being away from the 5100 Portal Area and Jordan River canyon.
- Select removal of some waste rock/tailings and cover to limit water infiltration.
- Underground disposal (i.e., in the mine workings - dropped in Step 2).
- Disposal in the 1963 Blowout (dropped in Step 2).
- Risk Manage the South Waste Rock Pile.



Off Site Disposal



Management in Place



Example Landfill Construction



Example Landfill Construction



Remedial Options Evaluated and Costed

Option	North Pile Waste Rock Pile		South Waste Rock Pile	Post-Remediation Condition (all include re-vegetation)
	High Sulphide and Highly Leachable Wastes	Low Sulphide Wastes		
A1	Reconfigure in-place and cover	Reconfigure in-place and cover	Risk manage	Retaining wall with composite cover
A2	Reconfigure in-place and cover	Reconfigure in-place and cover	Risk manage	Composite shotcrete cover
B1	Relocate to nearby constructed landfill	Relocate to nearby constructed landfill	Risk manage	Backfill to maintain access to portal
B2	Relocate to nearby constructed landfill	Cover in Situ	Risk manage	Backfill to maintain access to portal / retaining wall / composite geosynthetic cover
C1	Relocate to offsite permitted landfill	Relocate to offsite permitted landfill	Relocate to offsite permitted landfill	Backfill to maintain access to portal
C2	Relocate to offsite permitted landfill	Relocate to offsite permitted landfill	Risk manage	Backfill to maintain access to portal
C3	Relocate to offsite permitted landfill	Cover in Situ	Risk manage	Backfill to maintain access to portal / retaining wall / composite cover

Multiple Accounts Analysis – Summary of Results

ACCOUNTS	Option A1: In-Situ Regrading of Wastes	Option A2: In-situ Shotcrete Cover	Option B1: Move Entire North Pile to Nearby Constructed Landfill	Option B2: Move High Sulphide Wastes to Nearby Constructed Landfill	Option C1: Relocate North & South Pile to a Permitted Landfill	Option C2: Relocate North Pile to a Permitted Landfill	Option C3: Relocate High Sulphide Wastes to a Permitted Landfill
TECHNICAL	1.9	3.7	6.9	5.4	7.9	7.3	6.5
ECONOMICS	9.0	3.0	8.0	7.0	1.0	2.0	5.0
ENVIRO & PERMITTING	1.2	1.0	6.3	5.2	9.0	8.0	6.5
SOCIAL	Not Scored						
OVERALL SCORE	5.1	2.4	6.3	5.4	3.4	3.7	4.8

Candidate Options: Portal Discharge

- Portal Discharge (additional data required)

Risk manage (not likely).

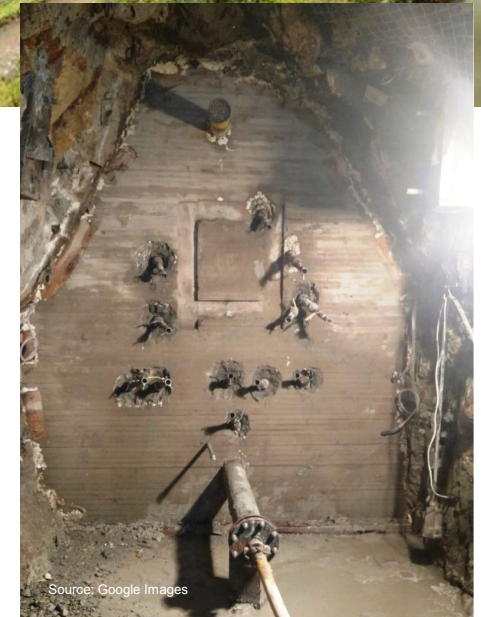
Passive or active treatment.

Marine disposal with diffusion pipes (dropped in Step 2).

- Portal Discharge (not in Teck control):

Plug portal and flood the mine to stop discharge.

Source control (stop water entering into the mine).



Other Factors to Consider in Closure Planning

Soil Management

- Land ownership
- Available borrow soil sources
- Landfill siting with groundwater studies
- Terrestrial risk assessment
- Archeological studies (new areas)
- Species at Risk and Migratory Birds

Water Management

- Water balance to support treatment options
- Source control (inflows from river)
- Geotechnical stability issues with portal and other mine workings
- Treatment options (active or passive)
- Bench and pilot tests to confirm water treatment methods

Implementation of a Remedial Option

Comprehensive Approach:

More data required to develop conceptual remedial options for the portal discharge.

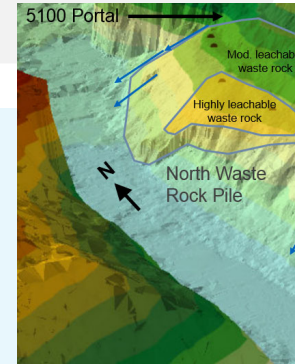
Addressing portal discharge may result in acceptable risk scenarios in the river and therefore waste rock management may not be required.

Construction sequencing is key due to steep terrain and difficult access.

Addressing waste rock in advance of the portal could result in inefficiencies.

Landowner, Regulator and COI engagement/acceptance is required.

Continued progress on Teck's and WFP's commitment to addressing Protocol 12 high-risk conditions. Acceptance from ENV on approach.



Conclusions: Remedial Options Analysis

- Preferred approach will likely include relocation of some waste rock and tailings to an onsite location.
- Ongoing discussions with the landowner regarding onsite management and potential locations, but nothing is confirmed .
- New Sunro Copper Ltd. holds the mine permit and maintains bulkheads and water flows – Teck doesn't not control this. Studies are ongoing for 2020.
- Constructability and timing must be considered to maintain a wide range of options for waste rock and portal discharges and lower costs (interim options to reducing loading could be considered).

Interim Options Considered

Any opportunistic improvements to salmon habitat?



Option	Key Advantages	Key Disadvantages
Geomembrane Covers	› Reduced net infiltration rates and less metal loading to the Jordan River › Risk reduction by removal of ecological and human terrestrial contact to contaminated soil › Potential improvement of water quality in Jordan River	› Increased safety concerns for persons trespassing at the › Fencing, gates, signage and inspections would likely be required in addition to the capital costs presented in this study › Damage by trespassers
Spray-On Liner	› Same as Geomembrane Covers › Increased geotechnical stability, degree not quantified	› Same as Geomembrane Covers; › Increased permeability compared to polyethylene geomembranes

Implementation of a Remedial Option

Gaps	Schedule
Flow Accretion Study and continued Water Sampling	2020
Continued portal investigations. (Geotech and hydrogeological investigation to assess closure option for portal)	2021 - 2022
Locating an area suitable for a landfill and collect baseline data (if this option is selected to proceed with)	2021
Assess archeological impacts from preferred option	Dependent on preferred option
Collect terrestrial samples to support post remediation risk assessment and changing geochemical conditions.	Dependent on preferred option
Interim cover to assess potential improvement in Jordan River (under consideration)	2021/22 (pending permit approvals)
Implementation of Remediation (Portal and Waste Rock Options)	5 years?

Sunro Mine

Remedial Options Analysis

Open Discussion



Teck